A Machine Learning Approach to Twitter Sentiment Analysis

CSCE 3602

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ABSTRACT

This project explores the use of machine learning for sentiment analysis on Twitter, aiming to classify tweets into positive, negative, or neutral categories. Social media platforms like Twitter generate massive amounts of unstructured text data, reflecting public sentiment on various topics such as products, politics, trends, and current events. By analyzing this data, organizations can gain insights into consumer opinions, market trends, and public discourse. Using a combination of traditional machine learning techniques and advanced deep learning models like BERT, this project evaluates different approaches to sentiment classification. Various datasets, such as Sentiment140, Twitter Sentiment, and Twitter Emotion Datasets, are utilized to train and test the models. The results of this analysis can be applied in brand monitoring, political analysis, and real-time market sentiment prediction, offering businesses and organizations valuable insights for decision-making.

CCS CONCEPTS

• Contextual Dependencies • Handling Sarcasm and Irony • Use of Emojis and Hashtags

KEYWORDS

Sentiment analysis, machine learning, natural language processing (NLP), Twitter, social media analysis, Naive Bayes, Support Vector Machines (SVM), BERT, LSTM, public opinion, brand monitoring, real-time market analysis.

1 Introduction

1.1 Social Media and the Rise of Public Opinion

In today's digital age, social media platforms like Twitter have become central hubs for public communication and opinion discussions. With millions of users actively posting their opinions on a wide range of topics, social media has become a key source of information reflecting public sentiment. Sentiment analysis, a subfield of natural language processing (NLP), is focused on interpreting and classifying the emotions expressed in these posts. Companies and organizations use sentiment analysis to measure public opinion, monitor brand perception, and understand customer sentiment. For instance, analyzing tweets can help businesses assess how consumers feel about their products or services, while political organizations can track public opinion on policies or candidates.

1.2 Challenges in Twitter Sentiment Analysis

Twitter data presents unique challenges due to its informal language, widespread use of slang, misspellings, abbreviations, and its strict 280-character limit per post. These characteristics make it difficult to apply traditional lexicon-based sentiment analysis, which often relies on pre-defined sentiment dictionaries. These lexicon-based approaches struggle to handle the context, sarcasm, and the dynamic use of language that is common on Twitter, limiting their accuracy and applicability.

1.3 Machine Learning Approaches to Sentiment Analysis

To overcome these limitations, machine learning-based approaches have been increasingly employed. These models are capable of learning from large datasets and are better suited to handle the complexities of informal language and evolving linguistic patterns. Algorithms such as Naive Bayes and Support Vector Machines (SVM) have been widely used for text classification tasks, including sentiment analysis. Additionally, more advanced deep learning architectures, such as Long Short-Term Memory (LSTM) networks and Bidirectional Encoder Representations from Transformers (BERT), have shown considerable improvements in capturing contextual dependencies and understanding the sentiment behind tweets.

1.4 Project Objectives and Applications

The primary goal of this project is to build a machine learning-based sentiment analysis model capable of accurately classifying tweets into three sentiment categories: positive, negative, and neutral. The project will encompass several key stages, including data collection, preprocessing, model development, and performance evaluation. By comparing the effectiveness of different machine learning models, this project will aim to determine the best approach for Twitter sentiment analysis.

The results of this study have significant practical applications. Sentiment analysis on Twitter can be used in brand monitoring, real-time market analysis, and public opinion tracking, providing valuable insights for businesses, political organizations, and market analysts. These insights allow decision-makers to react more effectively to emerging trends, customer opinions, and public discourse.

2 Literature Review

In order to better understand earlier attempts at tackling sentiment analysis of social media posts on Twitter/X, we had to review published research papers on the subject and we chose 4 papers that were published on this topic. All of the papers begin their approaches by preprocessing the text to remove any things that their models may struggle with, so we will refer to these preprocessing phases with letters from A to I just like what the authors of paper two did. The phases are: turning all text to its lowercase form (Phase A), replacing all links with 'URL' or removing them (Phase B), expanding grammatical contractions (Phase C), replacing emojis with 'EMOJI' or removing them (Phase D), replacing mentions with 'USER' or removing them (Phase E), removing non-alphabetic characters (Phase F), removing repeated letters (Phase G), removing stop words (Phase H), and lemmatization (Phase I). It is also worth noting that all of the papers use the same dataset that we intend to use, Sentiment140.

2.1 Paper 1: A Novel Machine Learning Approach for Classification of Emotion and Polarity in Sentiment140 Dataset

In this paper by Behera, Dash, and Roy, the authors decided to go for a classification strategy that is based on the Naive Bayes approach. They used the same dataset that we plan on using, which is Sentiment140. In their preprocessing phases, they dropped all columns but the text and target. For the target which had 2 unique values: 4 for positive and 0 for negative, they decided to replace the 4 with 1 to make this a binarized multinomial Naive Bayes classification problem. They then went through Phases A, B, F, G, and H. However, they have stated that rather than replacing any of the difficult-to-process words, they decided to remove them. They then decide to take a small 2000-row sample from the 1.6 million row data set and they do not state their training-testing split. Their model managed to reach an accuracy of 84.2% and a F-score of 85%. However, the lack of important details such as the testing-training split, the exact preprocessing steps, and the reason for the small sample size, make this paper quite weak.

2.2 Paper 2: Twitter Sentiment Analysis Using Different Machine Learning and Feature Extraction Techniques

The second paper, by Habib and Sultani, also decides to go for a traditional machine learning approach. However, it provides a much more in-depth explanation of the preprocessing and feature extraction techniques that they used. They propose using 4 different feature vector extraction techniques: Bag-of-words (BOW), Term frequency - Inverse document frequency (TFIDF), Word2vec, and Doc2vec. These are all different feature vector extraction techniques that will be used on the text after the preprocessing that they proposed which were named in the initial literature review paragraph. For the models that they used, they chose 3: Support vector machine, Naive Bayes, and Logistic Regression which they tested with 2 different preprocessing techniques and the 4 feature extraction techniques. They used a 80-20% split for training and testing, set the N-Gram size for the TFIDF and BOW to (1,2) and the feature vector size for the Word2vec and Doc2vec was 300. Which was admitted by the authors to be very low. They then ran their models for 2 preprocessing techniques, one that used all the phases and one that only used Phases C,D, and G. The Bag of Words feature vector extraction method in combination with the logistic regression model scored the best in both runs with an accuracy of 82.76% and 83.95% respectively.

2.3 Paper 3: Twitter Sentiment Analysis: A Comparison of Available Techniques and Services.

This paper, by Paolo Romeo, is similar to the previous paper that we discussed. However, it tries some neural models as opposed to statistical ones only as well as testing out ready-to-use services that attempt to tackle the same issue. The paper highlights common issues with sentiment analysis based on text and highlights ones not mentioned in other papers such as sarcasm and the importance of context. The author then goes through essentially the same preprocessing phases that were described earlier except lemmatization, which he stated reduced the number of unique words by 81.55% and the total number of words by 48,48%. For feature vector extraction, the author used CountVector from Sci-KitLearn and TFIDF for his statistical models and TensorFlow vectorization for his neural network models. For his statistical models, he used an 80-20% train-test split and decided to use Naive Bayes, Logistic regression, SVM, and Random Forests. For both CountVectorizer and TFIDF, Logistic regression had the highest accuracy of 76.31% and 76.64% respectively. For his neural network models, he used an artificial neural network approach, a convolutional neural network approach, a Bi-directional LSTM approach, and a hybrid Bi-LSTM-CNN approach. The author used a 60-20-20% train-test-validate split and managed to get a 72.48% accuracy rating for the ANN approach, 74.34% accuracy for the CNN approach, and 75.46% accuracy for the Bi-LSTM approach, and 74.39% accuracy for his hybrid approach. For the ready-to-use service models, he tried Amazon Comprehend and Google Cloud NLP API. Amazon Comprehend had a better result of 76.80% accuracy.

2.4 Paper 4: SENTIMENT ANALYSIS BASED ON MACHINE LEARNING METHODS ON TWITTER DATA USING oneAPI

In the final paper, by Sengul, Adem, and Yilmaz, the authors used Intel DevCloud for their text preprocessing, training, and testing in addition to the oneAPI framework to speed up the process. They do not state how they split their data. However, they do say that the preprocessing move included most of the phases discussed before except lemmatization. They decided to try four different approaches. The first approach was the Naive Bayes approach which got them an 80% accuracy rating, a linear SVM approach that got them a 82% accuracy rating, a logistic regression approach that got them 83% accuracy, and a Bi-LSTM-CNN-Maxpooling-Dense layer-Softmax layer approach that got them an accuracy of 85%.

3 Project Overview & Applications

For this project, we intend to create a machine-learning model that is competent at classifying text according to its sentiment. We plan on using the Sentiment140 dataset to train and test the model. We will focus on the target and text columns which we will test different preprocessing methods to see if we can improve the methodologies used by previous attempts at sentiment analysis especially the different phases introduced by paper 2. We will then try different feature vector extraction techniques as well as trying as many supervised learning models that are in the scope of the problem and course as possible in an attempt to bridge the gap between previous works on this subject in addition to trying to improve on them. Finally, we will choose the best result and proceed with it to the final milestone where it could be used for several applications such as product review analysis where you can categorize reviews according to their text, brand reputation management where you can see how a brand is doing according to recent social media posts about them, content recommendation systems where you can recommend certain products or content to people based on their comments or reviews on them, and public sentiment on news articles where you can see how the public is feeling about a certain news piece just from the comments on them.

4 Datasets

4.1 Sentiment140 dataset with 1.6 million tweets

4.1.1 Dataset Overview. It includes 1,600,000 labeled tweets. They were extracted using the twitter api . They are annotated such that 0 = negative, 2 = neutral, and 4 = positive.

4.1.2 Dataset Properties and Description.

1. Name: sentiment140
2. Source or link: kaggle https://www.kaggle.com/datasets/kazanova/sentiment140
3. Type of data: Tweets with sentiment labels (positive, negative, neutral).
4. Number of instances: 1.6 million tweets.
5. Number of features: 6 features (target, ids, date, flag, user, and text)
6. Availability: Open-source.

4.1.3 Relevance of the Dataset. Strongly relevant. It includes exactly what we need for our project, which is training data that contains the text of tweets and the target sentiment.

4.1.4 Dataset Limitations. It is biased towards the English language, neglecting other languages and cultures. Additionally, the tweets are dated 2009, which may not accurately reflect modern sentiment trends.

4.1.5 Example.
target: 0
ids: 1467811592
date: Mon Apr 06 22:20:03 PDT 2009
flag: NO_QUERY
user: mybirch
text: Need a hug

4.2 Twitter Sentiment Dataset 3 million labelled rows

4.2.1 Dataset Overview. It includes 3,000,000 labeled tweets. They are annotated such that 0 = negative, 1 = positive, and 2 = neutral..

4.2.2 Dataset Properties and Description.

1. Name: Twitter Sentiment
2. Source or link: kaggle https://www.kaggle.com/datasets/prkhrawsthi/twitter-sentiment-dataset-3-million-labelled-rows/data
3. Type of data: Tweets with sentiment labels (positive, negative, neutral).
4. Number of instances: 3.14 million tweets.
5. Number of features: 3 features (number, tweet, and sentiment)
6. Availability: Open-source.

4.2.3 Relevance of the Dataset. Strongly relevant. It includes exactly what we need for our project, which is training data that contains the text of tweets and the target sentiment.

4.2.4 Dataset Limitations. It includes some overlap with the sentiment140 dataset. In addition, it is largely dominated by the negative and positive sentiment, with negligible data on the neutral sentiment, which can skew model performance and make it less effective on datasets with varying sentiment distributions.

4.2.5 Example.
number: 800018
tweet: Dancing around the room in Pjs, jamming to my ipod. Getting dizzy. Well twitter, you asked!
sentiment: 1.0

4.3 Twitter Emotion Dataset

4.3.1 Dataset Overview. It includes 3,000,000 labeled tweets. They are annotated such that 0 = sadness, 1 = joy, 2 = love, 3 = anger, 4 = fear, and 5 = surprise.

4.3.2 Dataset Properties and Description.

1. Name: Twitter Emotion Dataset
2. Source or link: kaggle https://www.kaggle.com/datasets/adhamelkomy/twitter-emotion-dataset
3. Type of data: Tweets with sentiment labels (sadness, joy, love, anger, fear, and surprise).
4. Number of instances: 390,000 tweets.
5. Number of features: 3 features (number, text, and label)
6. Availability: Open-source.

4.3.3 Relevance of the Dataset. Strongly relevant. It includes exactly what we need for our project, which is training data that contains the text of tweets and the target sentiment.

4.3.4 Dataset Limitations. It is classified into five categories, which we need to further classify into the main three we are targeting. We also do not have information on the date or geography of the targeting and collection process of this data.

4.3.5 Example.
number: 3
text: i dont know i feel so lost
label: 0

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